

Introduction to Molecular Simulation

From Bits to Bonds — Computing Matter, One Atom at a Time

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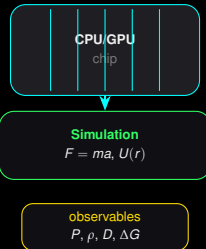
- ▷ 1. Introduction to Molecular Simulation
- ▷ 2. Statistical Mechanics
- ▷ 3. Molecular Representation on Computers
- ▷ 4. Accuracy and Approximation in Simulation
- ▷ 5. Time Scales in Molecular Simulation
- ▷ 6. Static vs Dynamic Molecular Systems
- ▷ 7. Molecular Dynamics
- ▷ 8. Simulation Data Analysis
- ▷ 9. Enhanced Sampling and Free-Energy Methods
- ▷ 10. Molecular Simulation Software
- ▷ 11. Hands-on LAMMPS Simulation
- ▷ 12. Introduction to GROMACS
- ▷ 13. Programming for Molecular Simulation

What is Simulation? — Control Over Numbers on a Chip

- ◇ **Simulation** = computable mirror of reality

Chip = Lab

Transistors → arithmetic → models → atoms → materials → life



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- ◇ **Control** → numbers you **ingrain** in silicon

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you control Δt , T , P !

CPU/GPU
chip

Simulation

$F = ma$, $U(r)$

observables

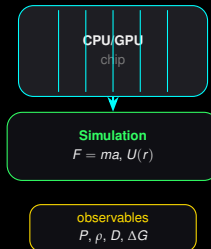
P , ρ , D , ΔG

What is Simulation? — Control Over Numbers on a Chip

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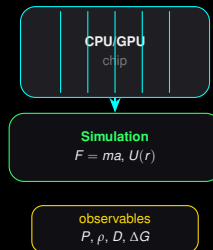


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- ◇ Input: `numbers` → Output: `properties`

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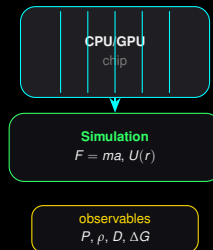


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- ◇ **Control** → numbers you **ingrain** in silicon
- ◇ Chip = deterministic universe you design
- ◇ Input: `numbers` → Output: `properties`
- ◇ **Why?** — experiment impossible / dangerous / expensive

Chip = Lab

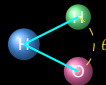
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Interpretation — keywords

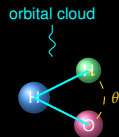
Simulation · mimic reality in code · **control** numbers → control physics · **chip** = lab · **reproducible** · **cheap** *what-if* · **not** magic — model quality = result quality

◇ **Orbitals** $\psi_{n\ell m} \rightarrow$ electron density



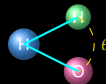
Wireframe $\leftrightarrow$ Ball-and-stick $\leftrightarrow$ Space-filling

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- ◇ Hydrogenic: $\hat{H}\psi = E\psi$, $\psi_{100} \propto e^{-r/a_0}$



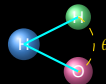
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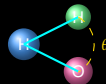
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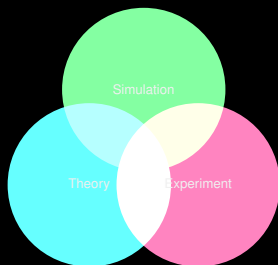
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- ◇ Models: **Quantum** (DFT) vs **Classical** (balls+springs)
- ◇ Precision ladder: **hydrogenic** $\rightarrow$ DFT $\rightarrow$ force field



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Interpretation — keywords

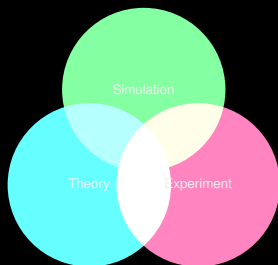
Modeling · choose resolution · **orbitals** = quantum truth · **atoms** = classical shortcut · **precision** costs · **ball-and-stick** = human eye model



◇ Theory → equations

Not a video game

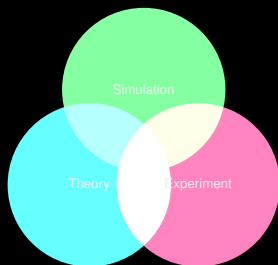
You don't watch molecules — you **measure** them: $g(r)$, MSD, ΔG .



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- ◇ **Experiment** → nature answers

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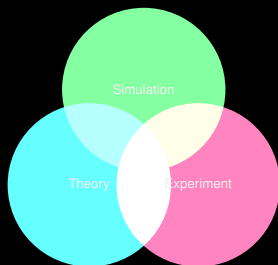
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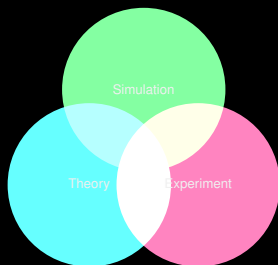
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- ◇ **Ball-and-stick:** atoms=spheres, bonds=sticks



Meshy

nano-banana — water waving

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- ◇ **Wireframe:** bonds only — topology



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- ◇ **Cartoon:** proteins → helices/sheets



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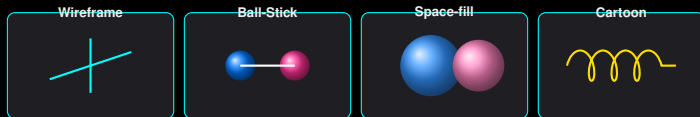
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- ◇ **Cartoon:** proteins → helices/sheets
- ◇ All derived from `coordinates + bonds`



Meshy

nano-banana — water waving



- ◇ **Structure:** $g(r)$, SASA, clustering

Success = Question + Model + Sampling

Bad question + good model = noise

Good question + bad model = **lie**

Drug binding → Materials → Energy → Bio

Simulation Objectives — What Are We After?

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- ◇ **Mechanism:** pathway, PMF
- ◇ **Design:** new molecule / material

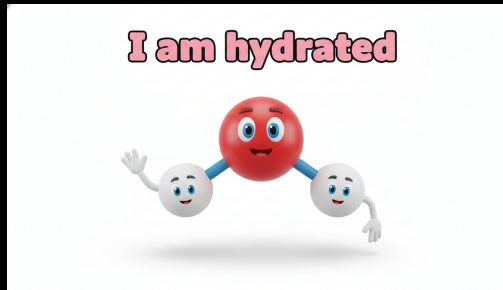
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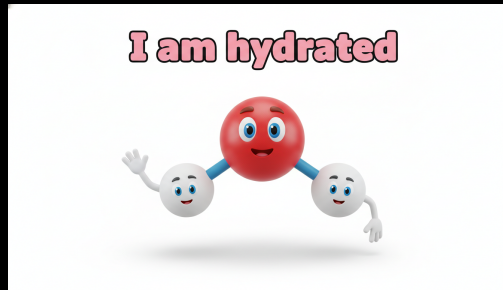
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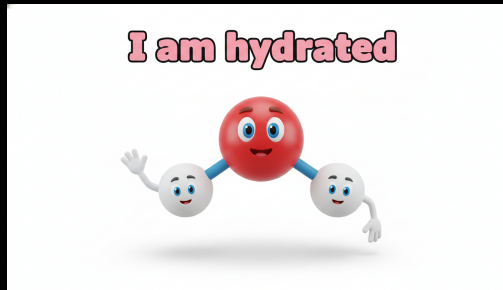
- ◇ Pharma: binding ΔG , **AlphaFold+MD**



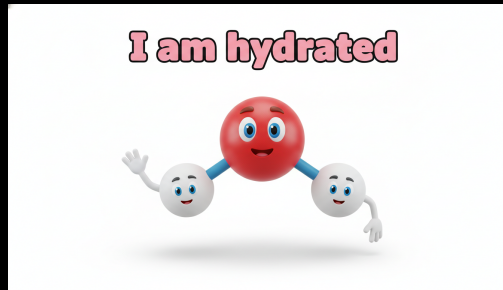
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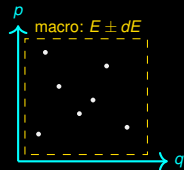
- ◇ Pharma: binding ΔG , **AlphaFold+MD**
- ◇ Materials: battery, polymer, catalyst
- ◇ Energy: CO2 capture, membranes
- ◇ Life: folding, crowding, rare events



Interpretation — keywords

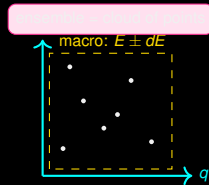
Objectives · **structure** / **dynamics** / **thermodynamics** · **mechanism** over movie · **design** is the test · **applications** drive approximations

- ◇ **Microstate:** $\Gamma = \{\mathbf{r}^N, \mathbf{p}^N\}$ — everything



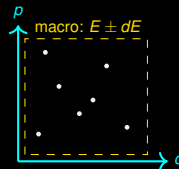
Microstate vs Macrostate

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- ◇ **Macrostate:** (N, V, E) or (N, P, T) — few numbers



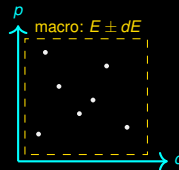
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- ◇ One macro $\leftarrow$ many micro: Ω microstates



Microstate vs Macrostate

- ◇ **Microstate:** $\Gamma = \{\mathbf{r}^N, \mathbf{p}^N\}$ — everything
- ◇ **Macrostate:** (N, V, E) or (N, P, T) — few numbers
- ◇ One macro $\leftarrow$ many micro: Ω microstates
- ◇ Phase space: $6N$ -dim, point = microstate

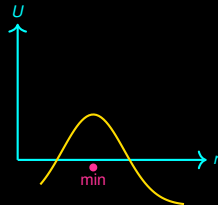


Interpretation — keywords

Micro · exact positions·momenta · **macro** · thermometer numbers · **phase space** · map of all micros · **one macro = many micros**

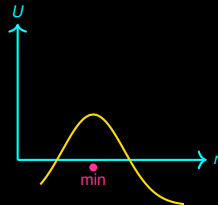
$$E = K + U = \sum_i \frac{\mathbf{p}_i^2}{2m_i} + U(\mathbf{r}^N) \quad \text{micro definition}$$

◇ $U(\mathbf{r}^N)$ = PES — see §3



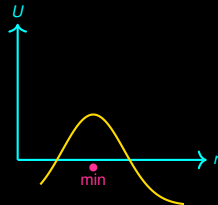
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- ◇ $U(\mathbf{r}^N)$ = PES — see §3
- ◇ Conserved in NVE: $dE/dt = 0$



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- ◇ $U(\mathbf{r}^N)$ = PES — see §3
- ◇ Conserved in NVE: $dE/dt = 0$
- ◇ Fluctuates in NVT: $\langle (\Delta E)^2 \rangle = k_B T^2 C_V$



$$S = k_B \ln \Omega \quad \text{and} \quad S = -k_B \sum_i p_i \ln p_i$$

◇ Ω = # microstates for macro

$$\Omega = 1$$

↓ $S \uparrow$

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- ◇ More ways $\rightarrow$ more entropy $\rightarrow$ more likely

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- ◇ Ω = # microstates for macro
- ◇ More ways $\rightarrow$ more entropy $\rightarrow$ more likely
- ◇ Disorder = **counting**, not mess

$$\Omega = 1$$

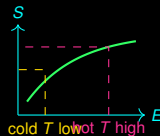
$\downarrow S \uparrow$

Temperature — Slope of Entropy

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{N,V}$$

hot = small slope

- ◇ Cold: S steep vs E

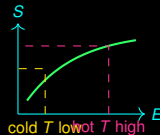


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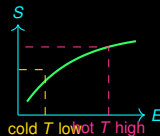


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- ◇ Cold: S steep vs E
- ◇ Hot: S flat vs E
- ◇ T = **exchange rate** of $E \leftrightarrow S$



Interpretation — keywords

Energy · conserved currency · **entropy** · count of ways · **temperature** · dS/dE · **hot** = flat $S(E)$

$$F = U - TS \quad G = H - TS \quad \min F, G \Leftrightarrow \text{stable}$$

- ◇ U wants order, S wants disorder



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- ◇ U wants order, S wants disorder
- ◇ F = compromise at T



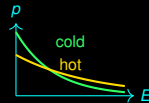
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- ◇ F = compromise at T
- ◇ Spontaneous $\Delta F < 0$



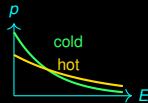
$$p_i = \frac{e^{-\beta E_i}}{Z}, \quad \beta = \frac{1}{k_B T}, \quad Z = \sum_i e^{-\beta E_i}$$

◇ Low $E \rightarrow$ high p



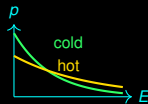
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- ◇ High $T \rightarrow$ flatter p



$$p_i = \frac{e^{-\beta E_i}}{Z}, \quad \beta = \frac{1}{k_B T}, \quad Z = \sum_i e^{-\beta E_i}$$

- ◇ Low $E \rightarrow$ high p
- ◇ High $T \rightarrow$ flatter p
- ◇ Z = partition function = normalization + **generator**



$$Z = \sum e^{-\beta E} \text{ (or } \int e^{-\beta H} d\Gamma) \quad F = -k_B T \ln Z$$

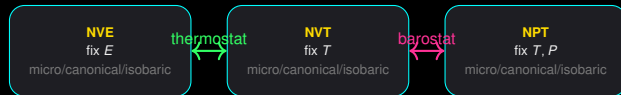
◇ $Z \rightarrow F \rightarrow P, S, C_V$ by derivatives

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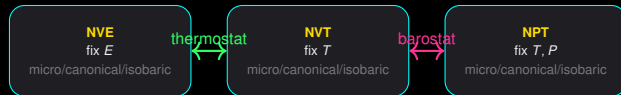
- ◇ $Z \rightarrow F \rightarrow P, S, C_V$ by derivatives
- ◇ Never need absolute Z , need **ratios** $\rightarrow$ **sampling**

Interpretation — keywords

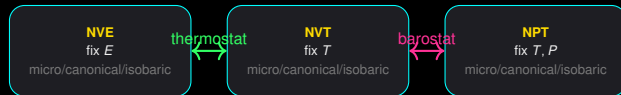
free energy · balance U vs TS · **Boltzmann** · low E wins · Z · sum that rules them all · **cold** = picky, **hot** = tolerant



- ◇ **NVE**: isolated, E conserved → MD default



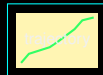
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- ◇ **NVE**: isolated, E conserved → MD default
- ◇ **NVT**: heat bath, T fixed → biology lab
- ◇ **NPT**: P + T fixed → real world

$$\langle A \rangle = \sum_i A_i p_i \approx \frac{1}{T} \int_0^T A(t) dt \quad \text{if ergodic}$$

- ◇ **Ergodic:** trajectory visits all with correct weight



ensemble (shaded)

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- ◇ Broken → trapped, **rare events** matter



ensemble (shaded)

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- ◇ **Ergodic**: trajectory visits all with correct weight
- ◇ Broken → trapped, **rare events** matter
- ◇ Solution: **enhanced sampling** (§9)



Interpretation — keywords

ensembles · what you fix · **NVE/NVT/NPT** · lab choice · **ergodic** · time = ensemble · **broken** → sample smarter

$$\mathbf{R} = \{\mathbf{r}_1, \dots, \mathbf{r}_N\}, \quad \mathbf{r}_i = (x_i, y_i, z_i) \in \mathbb{R}^3$$

- ◇ $3N$ numbers = **complete** classical description

```
ATOM 1 C 0.0 0.0 0.0
```

```
ATOM 2 O 1.2 0.0 0.0
```

```
ATOM 3 H -0.4 0.9 0.0
```

 array

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- ◇ File: `XYZ` / `PDB` / `GRO` → list
- ◇ Units: Å (most), nm (GROMACS)

```
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```

```
ATOM 2 O 1.2 0.0 0.0
```

```
ATOM 3 H -0.4 0.9 0.0
```

 array

$$\mathbf{R} = \{\mathbf{r}_1, \dots, \mathbf{r}_N\}, \quad \mathbf{r}_i = (x_i, y_i, z_i) \in \mathbb{R}^3$$

- ◇ $3N$ numbers = **complete** classical description
- ◇ File: `XYZ` / `PDB` / `GRO` → list
- ◇ Units: Å (most), nm (GROMACS)
- ◇ **Rigorous**: coordinates **are** the state

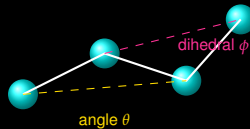
```
ATOM 1 C 0.0 0.0 0.0
```

```
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```

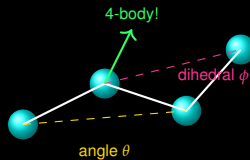
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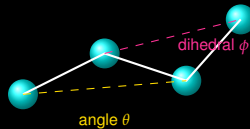
◇ **Bond:** $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$ · 2 atoms



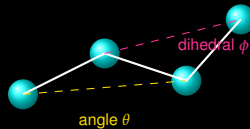
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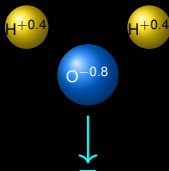
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- ◇ **Angle:** θ_{ijk} · 3 atoms
- ◇ **Dihedral:** ϕ_{ijkl} · 4 atoms, **periodic**
- ◇ Topology = **graph** (bonds = edges)



Interpretation — keywords

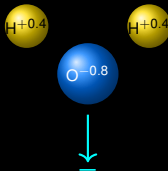
coordinates · $3N$ numbers = molecule · **bond** · distance · **angle** · 3 atoms · **dihedral** · 4 atoms = flexibility · **topology** · graph

◇ $m_i \rightarrow$ dynamics: $\mathbf{F} = m\mathbf{a}$

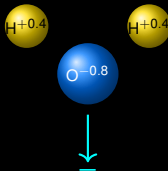


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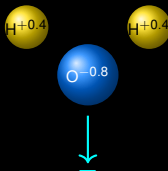
◇ $q_i \rightarrow$ electrostatics: $U_{\text{Coul}} = \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}$



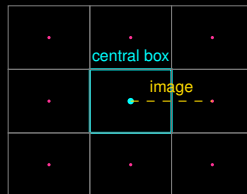
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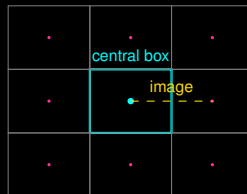
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- ◇ **Partial** $\neq$ formal: fitted to reproduce $\Phi(\mathbf{r})$
- ◇ Rigorous: charges are **parameters**, not observables



- ◇ Box: $\mathbf{L} = (L_x, L_y, L_z)$, triclinic: $\mathbf{h}$

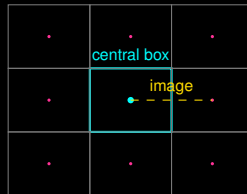


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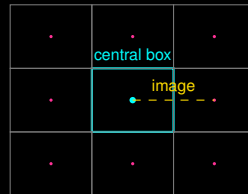
Simulation Box & Periodic Boundary Conditions

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$$U(\mathbf{R}) : \mathbb{R}^{3N} \rightarrow \mathbb{R} \quad \mathbf{F}_i = -\nabla_{\mathbf{r}_i} U$$

- ◇ $3N$ -dim surface $\rightarrow$ 2D cartoon is **projection**



Potential Energy Surface — The Mountain Range

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$$U = \underbrace{U_{\text{bond}} + U_{\text{angle}} + U_{\text{dihed}}}_{\text{bonded}} + \underbrace{U_{\text{LJ}} + U_{\text{Coul}}}_{\text{nonbonded}}$$

◇ Bond: $k_b(r - r_0)^2 \cdot \text{spring}$

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10^5 DFT points $\rightarrow U_{\text{FF}} \rightarrow 10^9$ steps

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- ◇ LJ: $4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$
- ◇ Many: AMBER, CHARMM, OPLS, ReaxFF, **ML potentials**

FF = Expensive Physics Compressed

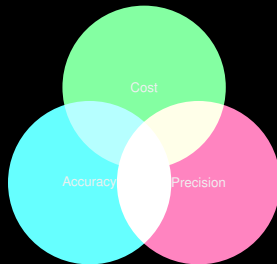
10^5 DFT points $\rightarrow U_{\text{FF}} \rightarrow 10^9$ steps

Interpretation — keywords

PES · energy landscape · **force** = slope · **force field** · cheap U · **bonded + nonbonded** · **parameterized** → not universal

Accuracy — What Does *Correct* Mean?

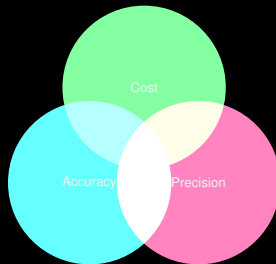
- ◇ **Accuracy** = closeness to **reality**



You can be precise and still wrong.

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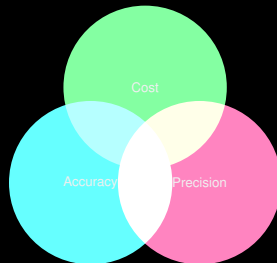
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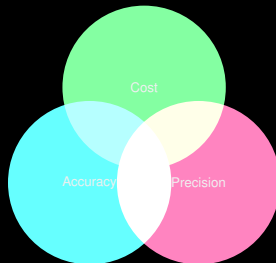
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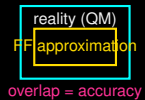
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- ◇ **Errors** stack: model + numerics + sampling

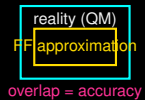


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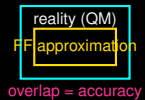
- ◊ Fixed topology → no bond breaking (except ReaxFF)



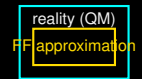
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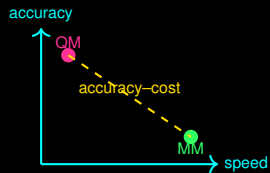


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- ◊ Pairwise $U_{LJ} \rightarrow$ no many-body
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- ◊ Harmonic bonds $\rightarrow$ no anharmonicity at large r

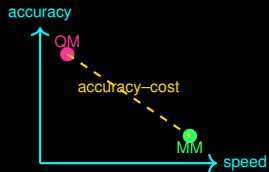


overlap = accuracy

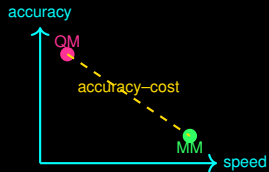
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- ◇ **Scale**: QM accuracy $\rightarrow$ MM speed



$$\min_{\{k_b, r_0, \epsilon, \sigma, q\}} \sum |U_{\text{FF}} - U_{\text{QM/exp}}|^2$$

- ◆ Fit to QM energies/forces or experimental ρ , ΔH_{vap}

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- ♦ Validate outside training set!

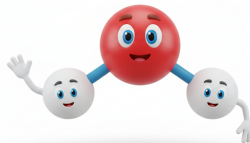
- Real $\in \mathbb{R} \rightarrow$ float $\in$ IEEE 754 (e.g., binary64: 1-11-52)

1 bit | 11 exp | 52 mantissa

↓round

$$0.1_{10} = 0.000110011..._2$$

I am hydrated



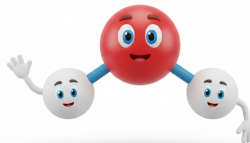
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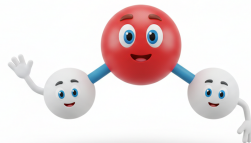
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I am hydrated



Numerical Errors — Computers Are Discrete

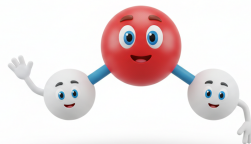
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- Accumulates: 10^9 steps $\times$ dt error

1 bit | 11 exp | 52 mantissa

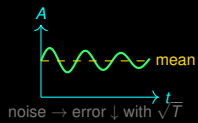
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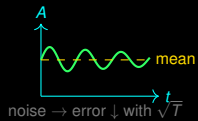
I am hydrated



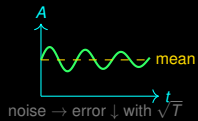
- Finite time $T \rightarrow \text{error} \propto 1/\sqrt{T/\tau}$



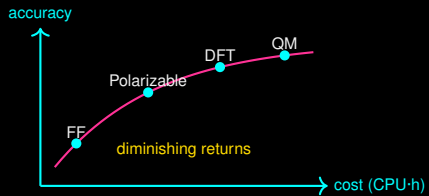
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- ◇ **Block averaging** $\rightarrow$ error bars



Accuracy–Cost Trade-off



Interpretation — keywords

accuracy · vs. Nature · **FF** = biggest error · **QM vs MM** · speed · **IEEE 754** · $0.1 + 0.2 \neq 0.3$ · **sampling** · $1/\sqrt{T}$ · **trade-off** · pay for truth

$$\mathbf{r}(t + dt) = \mathbf{r}(t) + \mathbf{v}dt + \frac{1}{2}\mathbf{a}dt^2 + \dots$$

- ◇ dt must resolve fastest vibration (bond ~ 10 fs)



The Timestep dt — Heartbeat of MD

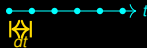
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- ◇ Rule: $dt \lesssim 1$ fs (flexible), 2 fs (constrained H)

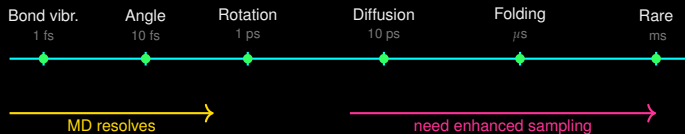


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- ◇ dt must resolve fastest vibration (bond ~ 10 fs)
- ◇ Rule: $dt \lesssim 1$ fs (flexible), 2 fs (constrained H)
- ◇ Too large $\rightarrow$ **explosion** (energy drift)



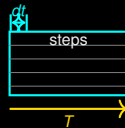
Hierarchy of Time



◇ **fs** → bonds, **ps** → water, **ns** → sidechains, μ **s** → folding, **ms–s** → rare

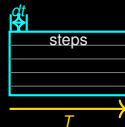
Simulation Length — $T = N_{\text{steps}} \times dt$

◇ $N_{\text{steps}} = T / dt$



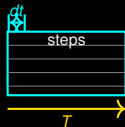
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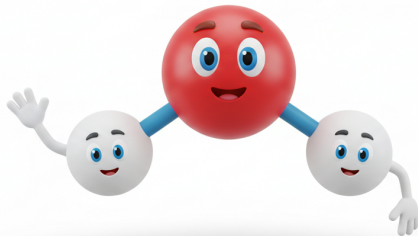


Simulation Length — $T = N_{\text{steps}} \times dt$

- ◇ $N_{\text{steps}} = T / dt$
- ◇ 100 ns at 2 fs $\rightarrow$ 50M steps
- ◇ Cost: 10^9 force calls = GPU-days



I am hydrated



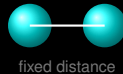
proteins — Russian doll of time

fs vibrations live inside ps water, inside μ s

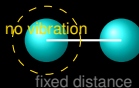
Interpretation — keywords

$dt \cdot 1\text{--}2 \text{ fs} \cdot \text{hierarchy} \cdot \text{fs} \rightarrow \text{ms} \cdot T = Ndt \cdot \text{steps} = \text{cost} \cdot \text{rare} \cdot \text{need tricks} \cdot \text{clock vs biology mismatch}$

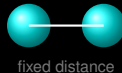
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- ◇ Where you look: **crystal, PDB, DFT optimum**

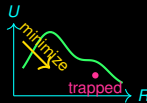


- ◇ **Static:** single $\mathbf{R}$, no T , no motion
- ◇ Where you look: **crystal**, **PDB**, **DFT optimum**
- ◇ Question: *what is stable?*



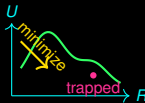
$$\min_{\mathbf{R}} U(\mathbf{R}) \Rightarrow \mathbf{F} = -\nabla U = 0, \text{ Hessian} > 0$$

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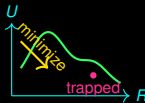
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DFT $\approx$ Exact for U , Still Static

No temperature $\rightarrow$ no entropy $\rightarrow$ free energy missing

- ◇ **Geom. opt.:** MM minimization (FF)
- ◇ **DFT:** QM minimization, *ab initio* U

DFT $\approx$ Exact for U , Still Static

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- ◇ **Geom. opt.:** MM minimization (FF)
- ◇ **DFT:** QM minimization, *ab initio* U
- ◇ Cost: DFT $\sim 10^3 \times$ MM

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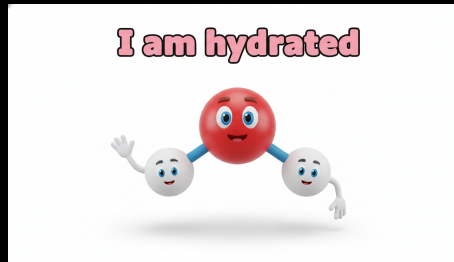
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- ◇ Still 0 K

DFT $\approx$ Exact for U , Still Static

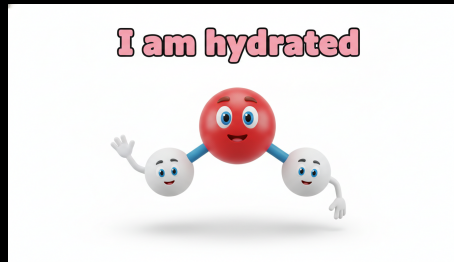
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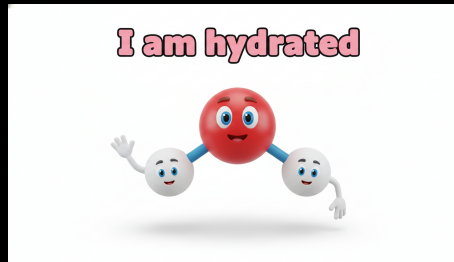
The Paradox — Why Proteins Fold Fast

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- ◇ Random search: 10^{27} yr



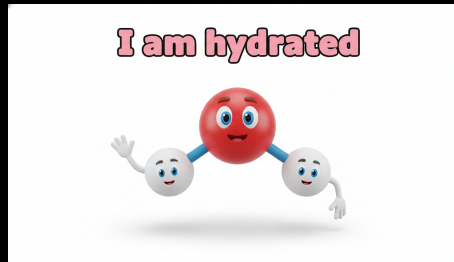
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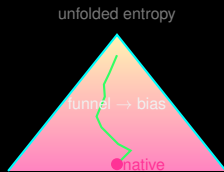
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The Paradox — Why Proteins Fold Fast

- ◇ Protein: 10^2 aa, $3^{100} \sim 10^{47}$ conformations
- ◇ Random search: 10^{27} yr
- ◇ Reality: ms–s → **Levinthal paradox**
- ◇ Resolution: **funnel**, not flat — downhill + guides





$$\Delta G = \Delta H - T\Delta S \quad p \propto e^{-\beta G}$$

◇ Min $U \neq \min F$ at $T > 0$

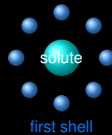
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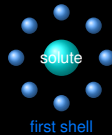
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- ◇ Min $U \neq$ min F at $T > 0$
- ◇ Entropy pushes to **broad** basins
- ◇ Solvent: **hydrophobic** = entropy of water

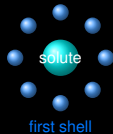
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- ◇ Explicit water: 10^4 molecules, PBC
- ◇ Implicit: GB/SA, continuum
- ◇ Rigorous: solvent = **free energy**, not just screening

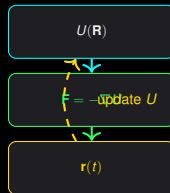


Interpretation — keywords

static · 0 K · min U · **Levinthal** · 10^{47} vs ms · **funnel** solves · **stability** = min F · **solvent** = entropy · **dynamic** = MD needed

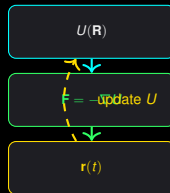
$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i = -\nabla_{\mathbf{r}_i} U(\mathbf{R}) \quad \mathbf{v}_i = \dot{\mathbf{r}}_i$$

◇ $U \rightarrow \text{force} \rightarrow \text{acceleration} \rightarrow \text{move}$



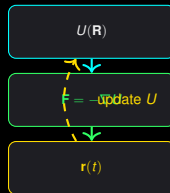
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- ◇ $U \rightarrow$ force $\rightarrow$ acceleration $\rightarrow$ move
- ◇ Need: **positions, velocities, forces**



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- ◇ $U \rightarrow$ force $\rightarrow$ acceleration $\rightarrow$ move
- ◇ Need: **positions**, **velocities**, **forces**
- ◇ Deterministic if U exact



$$\mathbf{r}(t + dt) = 2\mathbf{r}(t) - \mathbf{r}(t - dt) + \frac{\mathbf{F}(t)}{m} dt^2 + O(dt^4)$$

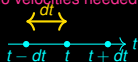
- ◇ **Verlet:** time-reversible, symplectic → stable



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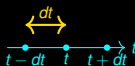
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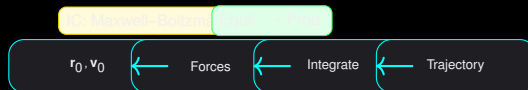
no velocities needed!



$$\mathbf{r}(t + dt) = 2\mathbf{r}(t) - \mathbf{r}(t - dt) + \frac{\mathbf{F}(t)}{m} dt^2 + O(dt^4)$$

- ◇ **Verlet:** time-reversible, symplectic → stable
- ◇ **Errors:** $O(dt^4)$ per step, $O(dt^2)$ global
- ◇ SHAKE/RATTLE → constrain H-bonds → larger dt

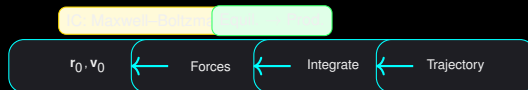




- ◇ **Initial conditions:** positions (PDB) + velocities ($\propto \sqrt{T}$)



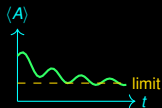
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- ◇ **Equilibration:** relax T, P , density plateau
- ◇ **Production:** N steps $\rightarrow$ **trajectory** + **sampling**

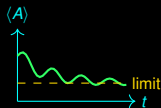
Convergence — When to Stop?

- ◇ **Convergence:** $\langle A \rangle$ stops drifting



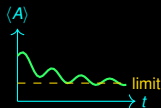
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- ◇ **Convergence:** $\langle A \rangle$ stops drifting
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Convergence — When to Stop?

- ◇ **Convergence:** $\langle A \rangle$ stops drifting
- ◇ Check: $\langle T \rangle$, RMSD plateau, block error
- ◇ **Longer $\neq$ converged** if τ huge



Interpretation — keywords

Newton · $F = -\nabla U$ · **Verlet** · reversible · **IC** · Maxwell · **trajectory** · equil. $\rightarrow$ prod. · **convergence** · plateau · **sampling** = statistics

◇ **Trajectory:** $\{\mathbf{R}(t_0), \mathbf{R}(t_1), \dots\}$ — DCD/XTC



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- ◇ **Size:** $N \times 3 \times N_{\text{frames}} \times 4$ bytes



- ◇ **Trajectory**: $\{\mathbf{R}(t_0), \mathbf{R}(t_1), \dots\}$ — DCD/XTC
- ◇ **Size**: $N \times 3 \times N_{\text{frames}} \times 4$ bytes
- ◇ **Analysis** = compress movie $\rightarrow$ numbers



RMSD

$$\text{RMSD}(t) = \sqrt{\frac{1}{N} \sum_i |\mathbf{r}_i(t) - \mathbf{r}_i^{\text{ref}}|^2}$$

drift = motion

RMSF

$$\text{RMSF}_i = \sqrt{\langle (\mathbf{r}_i - \langle \mathbf{r}_i \rangle)^2 \rangle}$$

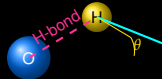
flexibility per residue

R_g

$$R_g^2 = \frac{1}{N} \sum_i |\mathbf{r}_i - \mathbf{r}_{\text{COM}}|^2$$

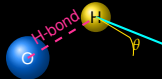
compactness

◇ H-bond: $d_{\text{O}\cdots\text{H}} < 3.5\text{\AA}$, $\angle > 150^\circ$



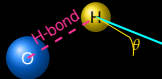
H-Bonds, Distances, Angles

- ◇ **H-bond:** $d_{\text{O}\cdots\text{H}} < 3.5\text{\AA}$, $\angle > 150^\circ$
- ◇ Distances/angles $\rightarrow$ **order parameters**



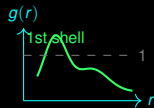
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- ◇ **H-bond:** $d_{\text{O}\cdots\text{H}} < 3.5\text{\AA}$, $\angle > 150^\circ$
- ◇ Distances/angles $\rightarrow$ **order parameters**
- ◇ **Define** before measuring



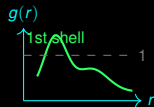
$$g(r) = \frac{\langle \rho(r) \rangle}{\rho_0} \quad \text{--- shell structure}$$

- ◇ First peak → coordination shell



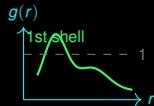
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- ◇ First peak $\rightarrow$ coordination shell
- ◇ $g(r) \rightarrow 1$ at long r
- ◇ Integrates to $n(r) = 4\pi\rho_0 \int_0^r g(s)s^2 ds$



- ◇ Group frames by RMSD → **clusters**



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- ◇ Representative = medoid



- ◇ Group frames by RMSD → **clusters**
- ◇ Representative = medoid
- ◇ **Why?** → reduce 10^6 frames → 5 states



- ◇ VMD/PyMOL/OVITO/ChimeraX → see artefacts

No Visualization = No Trust

Trajectory without watching = blind.

- ◇ VMD/PyMOL/OVITO/ChimeraX → see artefacts
- ◇ Color by **velocity**, **charge**, time

No Visualization = No Trust

Trajectory without watching = blind.

- ◇ VMD/PyMOL/OVITO/ChimeraX → see artefacts
- ◇ Color by **velocity**, **charge**, time
- ◇ **Titan** anecdote: 18,000 GPUs, 1 μ s/day → still **need** analysis

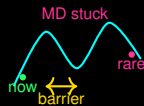
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Interpretation — keywords

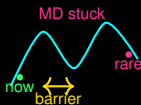
trajectory · movie · **RMSD** · drift · **RMSF** · flex · R_g · compact · **H-bond** · geometry · $g(r)$ · shells · **clustering** · 5 states · **visualize** · trust

- ◇ MD samples $\propto e^{-\beta U} \rightarrow$ rare events **never** seen



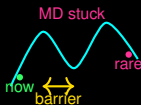
The Sampling Problem

- ◇ MD samples $\propto e^{-\beta U} \rightarrow$ rare events **never** seen
- ◇ Barrier $10 k_B T \rightarrow$ wait $e^{10} \sim 20k\times$ longer



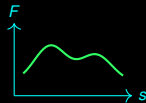
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- ◇ MD samples $\propto e^{-\beta U} \rightarrow$ rare events **never** seen
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- ◇ **Biology lives in rare events**



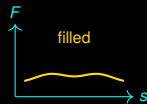
$$V_{\text{bias}}(s, t) = \sum_{t'} w e^{-(s-s(t'))^2 / 2\sigma^2}$$

- ◇ Deposit Gaussians along CV s



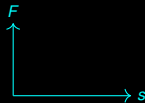
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- ◆ Deposit Gaussians along CV s
- ◆ Flood minima $\rightarrow$ escape
- ◆ **Free energy** = $-V_{\text{bias}}$ at convergence



Umbrella Sampling — Windows

- ◇ Harmonic umbrella per window: $w_i(s) = \frac{1}{2} k(s - s_i)^2$

Umbrella Sampling — Windows

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- ◇ Rigorous *PMF* via reweighting

- ◇ **Replica exchange:** hot replicas $\rightarrow$ cold via swap $P \propto e^{\Delta\beta\Delta U}$



- ◇ **Replica exchange:** hot replicas $\rightarrow$ cold via swap $P \propto e^{\Delta\beta\Delta U}$
- ◇ **Accelerated MD:** boost $U \rightarrow U + \Delta V$ when $U < \text{threshold}$



$$W(s) = -k_B T \ln P(s) \quad P(s) = \langle \delta(s - s(\mathbf{R})) \rangle$$

- ◇ PMF = free energy **along** CV

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- ◇ PMF = free energy **along** CV
- ◇ Minimum = stable, barrier = rate via Kramers

Interpretation — keywords

rare · barrier wins · **metadynamics** · fill wells · **umbrella** · windows → WHAM · **replica** · heat helps · **PMF** · $-kT \ln P$ · **free energy** = truth



- ♦ **LAMMPS**: materials, all-atom, coarse-grained, **you built it**



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- ♦ **LAMMPS**: materials, all-atom, coarse-grained, **you built it**
- ♦ **GROMACS**: biomolecules, fast, PME
- ♦ **OpenMM**: GPU, Python API, custom forces

Pick by System

Metals/polymers → LAMMPS
Biomembrane/protein → GROMACS/AMBER
Custom → OpenMM
Visualize → VMD/OVITO

All Need

Force field +++ **PBC** + thermostat + analysis

Interpretation — keywords

LAMMPS · materials · GROMACS · bio · OpenMM · GPU Python · VMD/OVITO · eyes · all · *U*+PBC+sampling

```
1 # Linux ( Xeons 24c + A100 sm_80 )
2 ./scripts/build-lammps.sh # most+gpu+kokkos
3 # Windows PowerShell
4 .\scripts\build-lammps.ps1
5 # Check
6 ./lammps/build/lmp -h | head
```

You Built 65 Packages

lmp -h shows GPU (mixed, sm_80) + KOKKOS (CUDA, AMPERE80) + 24 cores

```
1 units real
2 atom_style full
3 read_data water.data
4 pair_style lj/cut/coul/long 10
5 kspace_style ppm 1e-4
6 fix 1 all nvt temp 300 300 100
7 timestep 2.0
8 run 50000
```

◇ **Box:** `region + create_box` or `read_data`

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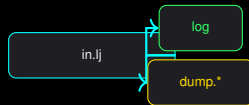
◇ **Atoms:** `create_atoms` or file

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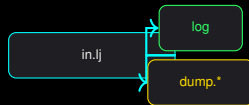
◇ **BC:** `boundary p p p`

◇ **dt:** 1–2 fs

◇ `lmp -in in.lj -log log.lammps`



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- ◇ **Log:** thermo (step, T , E , P)



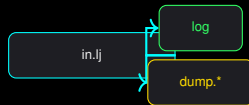
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- ◇ **Trajectory:** dump custom/xtc $\rightarrow$ VMD/OVITO



- ◇ `lmp -in in.lj -log log.lammps`
- ◇ **Log:** thermo (step, T , E , P)
- ◇ **Trajectory:** dump custom/xtc → VMD/OVITO
- ◇ **Analysis:** compute rdf, Python post



Interpretation — keywords

install · `build-lammps.sh` · **script** · 5 lines → run · **log** · T, E, P · **dump** · movie · **analysis** · your science

```
1 # Install (conda / apt)
2 conda install -c conda-forge gromacs
3 # Inputs
4 gmx pdb2gmx -f protein.pdb
5 gmx editconf -f conf.gro -o box.gro -c -d 1.0 -bt cubic
6 gmx solvate -cp box.gro -o solv.gro
```

◇ **Structure:** .gro/.pdb

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```

- ◇ **Structure:** .gro/.pdb
- ◇ **Topology:** topol.top (FF + bonds)
- ◇ **Parameters:** .mdp (dt , T , P)

`em.mdp → nvt.mdp → npt.mdp → md.mdp`



◇ **EM:** $\min U, F_{\max} < 1000 \text{ kJ/mol/nm}$

`em.mdp → nvt.mdp → npt.mdp → md.mdp`



- ◇ **EM:** $\min U, F_{\max} < 1000 \text{ kJ/mol/nm}$
- ◇ **NVT:** V-rescale thermostat, $dt = 2 \text{ fs}$, LINCS

`em.mdp → nvt.mdp → npt.mdp → md.mdp`



- ◇ **EM:** $\min U, F_{\max} < 1000 \text{ kJ/mol/nm}$
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- ◇ **Prod:** $50 \text{ ns} \rightarrow \text{traj.xtc} + \text{ener.edr}$

```
1 gmx rms -s topol.tpr -f traj.xtc -o rmsd.xvg  
2 gmx gyrate -s topol.tpr -f traj.xtc -o rg.xvg  
3 gmx rdf -s topol.tpr -f traj.xtc -o rdf.xvg
```

- ◇ Same as LAMMPS: RMSD, R_g , $g(r)$, H-bonds

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- ◇ Same as LAMMPS: RMSD, R_g , $g(r)$, H-bonds
- ◇ Different file names, same physics

Interpretation — keywords

GROMACS · bio focus · **files** · gro/top/mdp · **workflow** · min→NVT→NPT→prod · **analysis** · same $g(r)$, RMSD · **PMEs** = fast Coulomb

```
1 for d in */; do
2   mpirun -np 24 lmp -in in.lj
3   python analyse.py traj.xtc
4 done
```

◇ Linux: HPC speaks bash

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- ◇ **Linux:** HPC speaks bash
- ◇ **Automation:** loops, slurm, 'snakemake'
- ◇ **Reproducible:** script beats clicking

- ◇ NumPy/SciPy: arrays, FFT, linalg

```
1 import MDAnalysis as mda
2 import numpy as np
3 u = mda.Universe("topol.tpr", "traj.xtc")
4 for ts in u.trajectory:
5     pos = u.atoms.positions
6     # compute RDF, RMSD, ...
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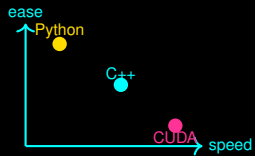
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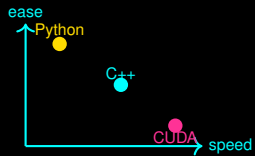
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- ◇ RDKit: cheminformatics

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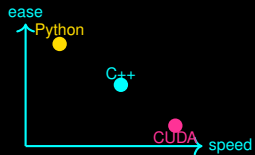
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- ◇ You: Python for glue, **CUDA** for custom kernels



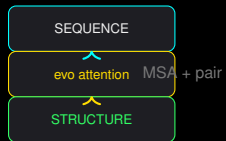
- ◇ LAMMPS/GROMACS = C++/CUDA/Fortran
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- ◇ `uv` → C++ → LAMMPS plugin



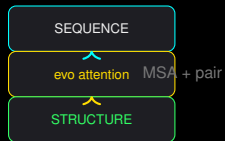
Interpretation — keywords

Bash · loop it · **Python** · NumPy/MDAnalysis · **C++/CUDA** · speed · **automation** · snakemake · **repro** = script

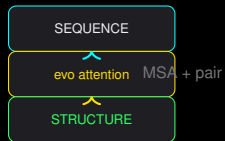
◇ Input: **sequence** → Output: **3D coordinates**



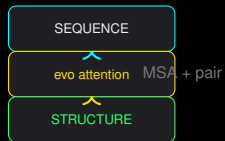
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- ◇ Attention over MSA → distances → structure module
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- ◇ **OpenFold** = open AlphaFold2



AlphaFold

One **static** structure
No dynamics, no F , no ensemble

MD

Ensemble of structures
 T , F , pathways, rates

AlphaFold = photo vs MD = movie

- ◇ **NIM**: containerized inference → AlphaFold3, OpenFold on GPUs



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- ◇ **AI + MD**: AI gives start, MD refines + samples **free energy**
- ◇ Rigorous: AI **does not** replace physics — it **accelerates** it



Interpretation — keywords

AlphaFold · sequence → structure · **photo** not movie · **MD** = ensemble+ F · **NIM** = GPU AI · **AI+MD** · predict then sample

```
1 git init
2 git add in.lj analyse.py
3 git commit -m "run 1: NVT 300K"
4 git tag v1.0
5 git log --oneline
```

◇ **Commit** = snapshot

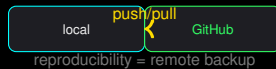
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- ◆ **Commit** = snapshot
- ◆ **Branch** = parallel try

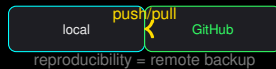
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5 git log --oneline
```

- ◇ **Commit** = snapshot
- ◇ **Branch** = parallel try
- ◇ **Tag** = publishable version

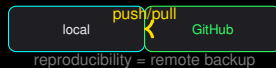
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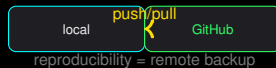
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- ◇ **Issues/PRs:** discuss, review
- ◇ **Actions:** CI — re-run every commit



◇ README → what + how + uv sync

Not Reproducible = Not Science

If someone can't re-run → it's a story, not a result.

- ◇ README → **what + how** + `uv sync`
- ◇ `pyproject.toml/uv.lock` → **exact versions**

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- ◇ README → **what + how + uv sync**
- ◇ `pyproject.toml/uv.lock` → **exact versions**
- ◇ **Scripts** + `lammps/build/lmp -h + seed`
- ◇ **Data**: `traj.xtc` + DOI (Zenodo)

Not Reproducible = Not Science

If someone can't re-run → it's a story, not a result.

Interpretation — keywords

Git · snapshot · **branch** · try · **GitHub** · share · **README** · how · **uv.lock** · versions · **repro** or it didn't happen



◇ **Prep:** crystal/PDB → `pdb2gmx` → `solvate` → ions

FF + Box + PBC = Lab

Your choices **are** the experiment.

- ◆ **Prep:** crystal/PDB $\rightarrow$ `pdb2gmx` $\rightarrow$ `solvate` $\rightarrow$ ions
- ◆ **Model:** choose FF (**parameterization** = fit)

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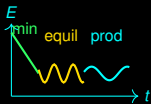
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- ◇ **Rigorous:** validate density, $g(r)$ before production

FF + Box + PBC = Lab

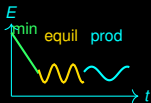
Your choices **are** the experiment.

- ◇ **Min:** remove clashes, $F_{\text{max}} \downarrow$



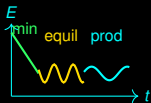
Minimization → Equilibration → Production

- ◇ **Min:** remove clashes, $F_{\max} \downarrow$
- ◇ **Equil:** NVT → NPT, T, P, ρ plateau



Minimization → Equilibration → Production

- ◇ **Min:** remove clashes, $F_{\max} \downarrow$
- ◇ **Equil:** NVT → NPT, T, P, ρ plateau
- ◇ **Prod:** **keep** trajectory, **log** T, E



◇ Analysis: RMSD, $g(r)$, clustering, *PMF*

Reproducibility

`git tag + uv.lock + seed + DOI` → someone can repeat 5 years later

- ◇ **Analysis:** RMSD, $g(r)$, clustering, *PMF*
- ◇ **Visualization:** OVITO/ChimeraX — **watch** first

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- ◇ **Analysis:** RMSD, $g(r)$, clustering, *PMF*
- ◇ **Visualization:** OVITO/ChimeraX — **watch** first
- ◇ **Interpretation:** numbers → **mechanism**

Reproducibility

`git tag + uv.lock + seed + DOI` → someone can repeat 5 years later

Interpretation — keywords

question · drives all · **prep** · solvate · **model** · FF · **min** → **equil** → **prod** · plateau · **traj** · movie · **analysis** · $g(r)$ · **repro** or non-science

Thank You — Simulate Responsibly

Code · Data · Reproducibility

```
git clone https://github.com/Shuvam-Banerji-Seal/introduction-to-molecular-simulation
```

```
uv sync && uv run python && ./lammmps/build/lmp -in in.lj
```